Subtract 00:07:30 from timestamps to find place in audio/video recording.

48
00:07:31.080 --> 00:07:42.010
Jamsheed Shorish: I'm not sure how many people we're going to have today. I know that a few people are off, Christina's off, Jacob is off. I'm not sure if there's a conflict for Zargum as well. It might be,

49
00:07:42.440 --> 00:07:52.719
Jamsheed Shorish: So we will see, but of course it's being recorded, and naturally it's a new topic, so it's good to kind of jump in, maybe it'll be more fun with a smaller group. But let's give it a few more minutes and see who's able to jump in.

50
00:07:53.540 --> 00:07:54.939
Krzysztof Paruch: Yeah, definitely, yeah.

51
00:07:55.240 --> 00:08:08.840
Krzysztof Paruch: This actually came… the idea for this topic came from conversations I had in Cannes at the Ethereum meetup. So we had a work, workshop, a working session, on token engineering in a breakout room.

52
00:08:09.060 --> 00:08:15.460
Krzysztof Paruch: And prediction markets are quite a hot topic at the moment, so many people come and want to talk about them, so…

53
00:08:15.710 --> 00:08:34.700
Krzysztof Paruch: we use this to motivate a discussion on the representation of claims, actually. So, like, everyone is talking about the different use cases, applications, and how markets can be designed. And also, in our conversation here, we've been predominantly speaking about market microstructure, and the trading layer, and the exchange layer.

54
00:08:34.860 --> 00:08:52.950
Krzysztof Paruch: But we haven't really put so much attention on the claims themselves, or the representation of claims themselves. And I looked into this over the last couple of weeks, and it's quite fascinating. So, like, it's actually a very important building block on the implementation side to understand

55
00:08:53.000 --> 00:09:00.749
Krzysztof Paruch: how those markets can operate, if you think about the proper representation of those claims. And there are

56
00:09:00.750 --> 00:09:14.439
Krzysztof Paruch: we will focus on this today, so, like, it's pretty nice to kind of, like, have this as a topic of conversation. As you say, like, it's kind of like a fresh start, like, you can jump in and come along pretty nicely, because it gives everyone a good chance. I will have a

57
00:09:14.440 --> 00:09:28.620
Krzysztof Paruch: a brief intro, even, on kind of, like, to motivate the topic. But it's… but it's in itself kind of like a new approach, right? A new, a new tangent that we're going to, to address today, looking at flame representations.

58
00:09:30.270 --> 00:09:30.910
Jamsheed Shorish: Great.

59
00:09:31.300 --> 00:09:40.249
Jamsheed Shorish: Well, cool, that's a great action that serves as a nice, a nice intro and lead-in all on its own, and a bit of a bridge from what you've been working on with the prediction market work beforehand, so…

60
00:09:40.510 --> 00:09:52.810
Jamsheed Shorish: Yeah, I think we're at 6 after David has joined. I think… yeah, why don't you go ahead and get started, because that way you get the time, and we can make sure that we reserve some time for discussion, perhaps at the end, if more people join.

61
00:09:53.800 --> 00:09:54.360
Krzysztof Paruch: Perfect.

62
00:09:54.970 --> 00:09:56.450
Krzysztof Paruch: Let's sort of… cool.

63
00:09:56.700 --> 00:10:14.009
Krzysztof Paruch: So let's talk about the conditional token framework, which is a core implementation primitive for prediction markets. As

mentioned before, we've talked mostly on the market layer, or the market microstructure, in those conversations here, but we will focus today

64
00:10:14.840 --> 00:10:22.559
Krzysztof Paruch: Primarily on the representation of claims, so the kind of assets that are representing those event contingent debts.

65
00:10:23.010 --> 00:10:30.249
Krzysztof Paruch: And how they are, represented in a technical environment, in the blockchain world.

66
00:10:30.400 --> 00:10:44.010
Krzysztof Paruch: And we will use the example of the conditional token framework, which was implemented or introduced by Knosis back in… I don't remember, but probably something around 2018, 2019.

67
00:10:44.440 --> 00:10:47.579
Krzysztof Paruch: And use it as the motivation for the conversation today.

68
00:10:47.950 --> 00:10:50.450
Krzysztof Paruch: So…

69
00:10:50.560 --> 00:11:02.779
Krzysztof Paruch: For everyone who was not here with us, let me motivate the topic by talking briefly about, binary production markets, so what it is, actually, that you're talking about here.

70
00:11:03.050 --> 00:11:16.149
Krzysztof Paruch: And we have those event contingent beds that are represented as claims, and they have this, their grounding, theoretically, in aerodiberal securities, so those are…

71
00:11:16.680 --> 00:11:26.580
Krzysztof Paruch: assets or securities, That pay out an amount, typically one, when an event occurs, and, nothing otherwise.

72
00:11:28.560 --> 00:11:33.289
Krzysztof Paruch: So, so this is what we… where we will start. In…

73
00:11:34.110 --> 00:11:39.350
Krzysztof Paruch: And the easiest way to introduce this is, looking at binary markets.

74
00:11:39.590 --> 00:11:43.859
Krzysztof Paruch: Where we'll… we will be looking at a situation where we have two claims.

75
00:11:44.180 --> 00:12:00.380
Krzysztof Paruch: And we have… those claims are exclusive and exhaustive, which means that we are basically… either one or the other happens, and there is nothing else that can happen, right? So, like, we have a yes or no question kind of thing, and we want to discuss the implications of this choice.

76
00:12:00.540 --> 00:12:03.039
Krzysztof Paruch: Relax this assumption a little bit.

77
00:12:03.350 --> 00:12:18.509
Krzysztof Paruch: and see how far we can get with this setup, if we, if we think about more sophisticated events, or even bet combinations. So, what we will find out today during this conversation is that, kind of like a binary

78
00:12:18.510 --> 00:12:23.410
Krzysztof Paruch: Setup is… is good for… to gain intuition and to motivate a topic.

79
00:12:23.490 --> 00:12:35.810
Krzysztof Paruch: But it doesn't get us that far, especially if we even think of very primitive representations of those things, as fungible tokens, ELC20s. So, like, we will not get very far with that.

80
00:12:35.860 --> 00:12:46.000
Krzysztof Paruch: What we will propose today is a different token standard that will have more richness in it, that will allow us to set up those more sophisticated claims.

81
00:12:46.150 --> 00:12:47.990
Krzysztof Paruch: And those conditioned bets.

82
00:12:48.580 --> 00:13:02.700
Krzysztof Paruch: And with that, we will also think about the market implications. So, like, how to design the markets, that we've, introduced before in previous conversations, and we will look into, kind of, like, how the conditional token framework even proposes

83
00:13:03.210 --> 00:13:10.279
Krzysztof Paruch: Particular market implementations that can handle those more sophisticated objects.

84
00:13:11.030 --> 00:13:11.810
Krzysztof Paruch: Okay.

85
00:13:12.210 --> 00:13:13.829
Krzysztof Paruch: So,

86
00:13:14.630 --> 00:13:21.589
Krzysztof Paruch: What I have here, this comes from the draft paper that I have shared, I think, a few months back.

87
00:13:22.440 --> 00:13:34.020
Krzysztof Paruch: just basically, again, motivation, even further motivation of, like, the state space that we're talking about here. So, we have this event space, E&A, where we have, kind of,

88
00:13:34.020 --> 00:13:43.410
Krzysztof Paruch: E, the event, which is the question that we are asking, and R, A, are all the outcomes, or the claims that we are…

89
00:13:43.410 --> 00:13:46.990
Krzysztof Paruch: And allowing for this, answer of the question to be.

90
00:13:46.990 --> 00:13:59.680
Krzysztof Paruch: And then we're talking about, kind of, like, some time representation, we're talking about the outstanding shares that are corresponding to those outcomes, and we're talking about the collateral or liquidity of those assets.

91
00:14:00.250 --> 00:14:03.570
Krzysztof Paruch: So we have this state-based representation. This is

the object that we're talking about.

92
00:14:04.360 --> 00:14:19.880
Krzysztof Paruch: And from this S, we can derive the instantaneous price vector P, which would be basically looking at those outstanding shares, and then trying to find relationships between them, and seeing how much shares are outstanding, how shares are priced.

93
00:14:20.260 --> 00:14:23.910
Krzysztof Paruch: accordingly, and, what…

94
00:14:24.370 --> 00:14:37.560
Krzysztof Paruch: if this is normalized, this then implies even a market probability. So… so we have what we're looking for, right? Like, here in this case, we are designing a market to reveal information, rather than we have assets.

95
00:14:37.640 --> 00:14:51.700
Krzysztof Paruch: that we want to facilitate the exchange for by designing the market. Here, we are generating synthetic assets because we are more interested in the trading behavior and the information revelation through actions of agents.

96
00:14:51.700 --> 00:14:58.599
Krzysztof Paruch: Rather than, kind of like, just a simple processural, capability of Exchange.

97
00:14:59.520 --> 00:15:16.000
Krzysztof Paruch: Okay, and what we obviously need in those scenarios always is, the possibility to resolve the outcome. So, like, we have, once we know the realized outcomes, like, we know which of those events actually has happened, so, like, what is the true answer for the question that we had?

98
00:15:16.500 --> 00:15:25.610
Krzysztof Paruch: Which is typically provided by some neutral entity referred to as an oracle, and then we can resolve the market, and we can pay out outstanding claims.

99
00:15:25.610 --> 00:15:42.859
Krzysztof Paruch: Which works in the following fashion. So we have this definition that the resolution, each share, which is outstanding,

obtains the payout P-bar if the event occurs, and zero otherwise, and P bar is typically 1. So, like, this is what we…

100
00:15:42.860 --> 00:15:58.220
Krzysztof Paruch: what we had, but more generally, PBAR is basically kind of, like, the cost of, like, a full set of claims, so that you can generalize this, right? Like, in a scenario where you pay out one, it would mean that for each unit of collateral, you are minting a full set of shares?

101
00:15:58.220 --> 00:16:03.630
Krzysztof Paruch: But if you, kind of, like, the payout is B-barred, then, like, you have to pay exactly this amount of collateral.

102
00:16:03.660 --> 00:16:06.240
Krzysztof Paruch: to achieve the same thing, okay?

103
00:16:06.400 --> 00:16:11.810
Krzysztof Paruch: So this is what's happening here, and this is what we're talking about.

104
00:16:12.020 --> 00:16:35.390
Krzysztof Paruch: So, how does this look in a binary event? We have this unit of collateral, which is 1, and what we can always do is we can decompose it into two complementary contingent claims, right? Because we have this exclusive, exhaustive relationship, so we can say, okay, for each asset that you pay, I can give you one yes and one no, because either of those has to occur, so, like.

105
00:16:35.390 --> 00:16:56.959
Krzysztof Paruch: In sum, or, like, in aggregation, they are actually worth the one collateral, so, like, they are always worth one, and since they are conditioned on some future event, their prices might be volatile, or might fluctuate in the future, but they will always add up to 1, and at the end, they will probably even converge towards one or zero, and this will

106
00:16:56.960 --> 00:17:00.209
Krzysztof Paruch: Basically, show, kind of, like, certainty, or, like.

107
00:17:00.240 --> 00:17:04.210
Krzysztof Paruch: Yeah, probabilistic certainty in the event to occur.

108
00:17:04.690 --> 00:17:16.700
Krzysztof Paruch: So, so this is what's happening here, and, you set up things like the primary market and the secondary market for this, where in the primary market, we just create those claims.

109
00:17:16.910 --> 00:17:22.579
Krzysztof Paruch: And in the secondary market, we allow for trade to happen.

110
00:17:22.579 --> 00:17:28.170
Krzysztof Paruch: between yes and no, and then we have multiple options how we can set it up. So, like, we can, for example, just

111
00:17:28.170 --> 00:17:51.719
Krzysztof Paruch: represent those things, and a yes asset as kind of, like, a fungible token, like in your C20, for example, and then we can say this token is quoted, so, like, this is tradable, and we will have price discovery for yes, and then we will have this, no arbitrage condition, or, like, some kind of, like, relationship where we will just dictate no to be 1 minus PES, right?

112
00:17:51.720 --> 00:18:09.389
Krzysztof Paruch: Which is the simplest thing. So, like, we can basically just mint only one token, and then we can trade it, and then we can allow the exchange for the other token by just simply deriving the price from the price of the other one, rather than kind of, like, having an actual market for it. We could…

113
00:18:09.490 --> 00:18:27.770
Krzysztof Paruch: obviously, also kind of, like, in some redundant approach, also set up two marketplaces to allow trading for yes and for no separately, but then we would, rely on market participants, arbitrageurs in this context, to ensure price adjustment. So, like, kind of, like, yeah.

114
00:18:27.910 --> 00:18:33.430
Krzysztof Paruch: Adjust prices and then synchronize prices in between those two markets.

115
00:18:34.680 --> 00:18:47.160
Krzysztof Paruch: Or, what we also could do is we could actually set

up an AMM, so, like, like a decentralized exchange, kind of thing, where we have, both assets pooled

116
00:18:47.160 --> 00:18:52.130
Krzysztof Paruch: against each other. So we are not trading against Numeraire, we are not trying to find

117
00:18:52.130 --> 00:19:07.260
Krzysztof Paruch: to have price discovery against a reference currency, we are actually pooling those assets against each other, and just allowing price discovery through adjustment of positions, or adjustment of inventories in this context.

118
00:19:08.500 --> 00:19:14.169
Krzysztof Paruch: What, we… Like, after this motivation, what we arrive at here is

119
00:19:14.460 --> 00:19:23.200
Krzysztof Paruch: It's basically this conceptually strong distinction between the claim representation itself and the settlement layer that we have. So, like, how do we represent positions?

120
00:19:23.570 --> 00:19:31.759
Krzysztof Paruch: And, up until now, we thought, okay, year C20s are fine, it's good enough, like, why not? We just kind of, like, need something fungible to trade with.

121
00:19:32.000 --> 00:19:38.329
Krzysztof Paruch: And the market layer, where we have this claim, exchange, and price-finding layer that can be enabled

122
00:19:38.510 --> 00:19:46.409
Krzysztof Paruch: In a different fashion. So, things that we actually addressed in previous conversations. We don't want to talk much about this today.

123
00:19:46.890 --> 00:20:02.330
Krzysztof Paruch: Okay, so let's talk now about limitations of this. This approach is obviously intuitive, but has very practical limitations. We've been so far only introducing on, like, a binary case, like, very reduced scope.

124
00:20:02.330 --> 00:20:14.340
Krzysztof Paruch: But if we talk about more sophisticated or, events or bad combinations, this would be impractical, and the complexity would be… become very quickly overwhelming, so we don't… we will…

125
00:20:14.540 --> 00:20:17.160
Krzysztof Paruch: I quickly see that we don't want to go this way.

126
00:20:17.180 --> 00:20:31.809
Krzysztof Paruch: So, let's talk about an event that emits N mutually exclusive and exhaustive outcomes, so we don't have any yes or no questions, but we have a set of events, of N events.

127
00:20:31.810 --> 00:20:38.290
Krzysztof Paruch: That, are, again, exclusive and exhaustive, where we say, okay, one of them will happen, and only one will happen.

128
00:20:39.330 --> 00:20:43.579
Krzysztof Paruch: And we have the same condition that,

129
00:20:43.700 --> 00:20:48.809
Krzysztof Paruch: they, pay out to one, or pay out to PBAR, actually.

130
00:20:50.400 --> 00:20:58.850
Krzysztof Paruch: What obviously follows is that we can decompose this unit of collateral into a set of N securities, or security claims.

131
00:20:59.190 --> 00:21:07.449
Krzysztof Paruch: And, we have the same condition that the claim pays us 1 if this specific one event occurs and 0 otherwise.

132
00:21:08.100 --> 00:21:24.850
Krzysztof Paruch: And we have the same thing, so no arbitrage, and so on and so on. This allows us to generalize immediately to more complicated markets. So we have categorical markets, we have ranked outcomes, color ranges, or mutually exclusive scenario sets. So we have richness in the outcome space.

133
00:21:24.880 --> 00:21:34.509
Krzysztof Paruch: But, the question now becomes, like, how do we want

to represent this? Do we want to represent each of those outcomes in S&RC20? Do we want to, kind of.

134
00:21:34.510 --> 00:21:45.800
Krzysztof Paruch: set up a multi-dimensional AMM, like a balancer, for example, where we would be able to trade all of those assets against each other and price them. What's happening here? So, like, how do we deal with that, right?

135
00:21:47.210 --> 00:22:04.119
Krzysztof Paruch: Here, it's very interesting to look on, like, the initial motivation. So, theoretically, this is grounded in Hansen's seminal paper from 2003, where he just, like, proposes logarithmic market scoring rules and combinatorial market design.

136
00:22:04.250 --> 00:22:11.250
Krzysztof Paruch: And, and this is the theoretical grounding, and we will see that this is one way to look at it, through kind of, like, an

137
00:22:11.370 --> 00:22:13.789
Krzysztof Paruch: Like, the economics of it.

138
00:22:13.960 --> 00:22:30.170
Krzysztof Paruch: But remember, for today, we are really looking at the implementation. So, like, we really want to see how those assets can be represented, and how can we even realize this on the blockchain infrastructure. So, we will find that although the

139
00:22:30.170 --> 00:22:41.549
Krzysztof Paruch: solutions proposed in this paper are very appealing, they are solving the issue, or approaching the issue on a different level. They are approaching this issue on an economic level, and not so much on the implementation on the technical level.

140
00:22:44.630 --> 00:23:05.250
Krzysztof Paruch: So, let's look at Hanson, who is, proposing multiple linked events. So, like, we have a set of M events, E1 to EM, and they have a joint outcomes space omega, which is the product of all of those A's. And traders now might be interested in combinations of outcomes.

141
00:23:05.330 --> 00:23:18.830

Krzysztof Paruch: So, for example, they are saying, okay, I'm interested in A and B occurring, I'm interested in A or B occurring, or I'm interested in A occurring if B occurs as well, right?

142
00:23:18.830 --> 00:23:28.230
Krzysztof Paruch: So, again, more sophisticated bets and more sophisticated combinations, and we see that, like, the complexity of this becomes very, very…

143
00:23:28.230 --> 00:23:46.209
Krzysztof Paruch: very high quickly, so the obvious answer is that this doesn't work so well with one ESC20 per outcome design, because we get, in the worst case, to, a very strong or very high size of the set.

144
00:23:46.330 --> 00:23:50.910
Krzysztof Paruch: Which is 2 to the power of, of omega, you know.

145
00:23:51.830 --> 00:23:58.600
Krzysztof Paruch: So this is operationally very inefficient, right? So what we are looking now is, okay,

146
00:23:58.690 --> 00:24:02.560
Krzysztof Paruch: We're looking for a different solution. ELC20s, seem to kind of, like.

147
00:24:02.610 --> 00:24:07.289
Krzysztof Paruch: be not good enough. We're looking for a system that, allows you

148
00:24:07.290 --> 00:24:14.230
Krzysztof Paruch: To, have one collateral asset that can back many claims, or multiple claims, or complicated claims.

149
00:24:14.230 --> 00:24:29.809
Krzysztof Paruch: There can be different claim types that can co-exist in one contract, so we don't want to, you know, set up, like, 50 contracts just to represent all of the outcomes and then trade combinations of them. We want to have one claim, one asset that is representing all of this.

150

00:24:29.810 --> 00:24:42.360
Krzysztof Paruch: We want to have the feasibility to recombine and split claims, so we want to have within the standard, or within the structure, already some of the combinations

151
00:24:42.360 --> 00:24:58.329
Krzysztof Paruch: that are mentioned before, or the operations that have been mentioned before, like creating claims and redeeming them, or combining them for one unit of collateral, we want this to be facilitated by the standard or by the contract itself.

152
00:24:58.510 --> 00:25:14.380
Krzysztof Paruch: And we also want to look at joint and conditional outcomes that can be represented efficiently. And this is exactly the motivation for this conditional token framework, which is basically elevating the ERC20 logic into an ERC1155.

153
00:25:14.540 --> 00:25:29.059
Krzysztof Paruch: And that's kind of the summary of what we're gonna talk about for the next 40 minutes, but we will be obviously looking into the details of how this is exactly set up, and what are the implications, and what,

154
00:25:29.370 --> 00:25:32.529
Krzysztof Paruch: And how does this work step-by-step if we look into the code?

155
00:25:32.840 --> 00:25:37.720
Krzysztof Paruch: Now would be a good time to make a brief break, and maybe allow some comments or questions.

156
00:25:37.830 --> 00:25:43.019
Krzysztof Paruch: In the meantime, I will have to do something for one minute, but I will stay connected and listen, and I will just be muted.

157
00:25:57.760 --> 00:26:01.780
Jamsheed Shorish: I guess one of the questions that comes up, and more… more of something for… for…

158
00:26:02.350 --> 00:26:20.919
Jamsheed Shorish: Future thoughts, perhaps, not just on

implementation, but also upon use cases, is the extent to which individuals are banking on or are seeking an instrument that can, in fact, allow them to place multiple simultaneous bets on potentially non-mutually exclusive events occurring for different types of correlation.

159
00:26:20.920 --> 00:26:28.680
Jamsheed Shorish: there, we're kind of, like, jumping back, just to go back to traditional markets, the idea of a mutual fund or an ETF, something that actually is

160
00:26:28.760 --> 00:26:47.539
Jamsheed Shorish: looking for particularly targeted sectors, but is… and therefore is accepting a certain type of, you know, systemic risk that goes across the sector, but then looks for idiosyncratic changes, bets, let's call them, on the performance of different members of that sector, different firms, or different types of

161
00:26:47.540 --> 00:26:51.330
Jamsheed Shorish: Even of instrument balances that are within a single mutual fund, for example.

162
00:26:51.330 --> 00:27:01.020
Jamsheed Shorish: Are we thinking about that as what this is? Is there still the information revelation component that is really what drives the prediction market?

163
00:27:01.020 --> 00:27:16.959
Jamsheed Shorish: Inside this framework, or are we evolving, in a sense, into, essentially kind of risk management of portfolios now, of what might be construed as, now loosely in quotations, you know, sort of sector risk smoothing?

164
00:27:18.320 --> 00:27:38.280
Krzysztof Paruch: I think that, the intuition for the conversation in the first place is to, just, achieve technical feasibility for the implementation, right? So, like, this is where, kind of like, I come from. But, your questions are, obviously completely legit. So the question… the answer is,

165
00:27:39.540 --> 00:27:50.329
Krzysztof Paruch: what is it… is it that I kind of, like, solved with this approach? Is it only that I can actually implement it, and I can

run the system? Because now I have

166
00:27:50.330 --> 00:28:04.699
Krzysztof Paruch: overcome the theoretic… the technical complications and computational limitations to just enable the system, or do I also buy something else with this approach, such that I actually can

167
00:28:04.970 --> 00:28:14.799
Krzysztof Paruch: Build more sophisticated use cases and look at the benefits economically as well, in the sense that, as you have mentioned.

168
00:28:14.820 --> 00:28:28.170
Krzysztof Paruch: More, information revelation, and, and, yeah, and, and these kind of things, that, that helped me to, to build more sophisticated products.

169
00:28:31.860 --> 00:28:39.550
Jamsheed Shorish: Okay, so really the focus at this stage is really, can ERC 1155 sort of replicate the ability to build a,

170
00:28:39.720 --> 00:28:59.569
Jamsheed Shorish: Yeah, a portfolio solution, let's say, of a variety of different assets, each of which asset has a particularly clearly distinguished type of outcome that can be verified, let's say, with an Oracle. As long as we kind of keep it scoped to that, then I think that makes sense, and we don't have to generalize to, like, you know, why are people using it, how, you know, for what use cases are there at this point, then? Does that make sense?

171
00:28:59.990 --> 00:29:17.269
Krzysztof Paruch: Right, this makes sense. Like, to me, the answer is, like, you have a blueprint, or, like, an explanation, like, Hanson introduced this 2003, 2007, and then people were wondering, how can I actually build it, and then Ignosis came along and said, okay, let's introduce a framework, technically, that, exactly maps

172
00:29:17.270 --> 00:29:23.320
Krzysztof Paruch: the implementation standard to what the paper is proposing. So, like, how can I technically

173
00:29:23.810 --> 00:29:37.820

Krzysztof Paruch: build this? How can I solve this, this economic system? And, and the answer is, yeah, the CTF is exactly the system that we were looking for, because it

174
00:29:37.960 --> 00:29:41.630
Krzysztof Paruch: Allows, kind of like, this market to function in the way that it's described.

175
00:29:43.370 --> 00:30:00.890
Krzysztof Paruch: And potentially, but I have not looked into this, we can ask follow-up questions. Like, we can ask the question of, okay, now that we've got here, what can we actually build, right? So, like, to your point, I'm confident, or I'm happy at the stage that, like, okay, there is something that this,

176
00:30:00.890 --> 00:30:09.090
Krzysztof Paruch: building, or facilitating the implementation of it, but I have not really questioned, what else can we build, or, like, what is the…

177
00:30:09.340 --> 00:30:10.899
Krzysztof Paruch: What can we build on top?

178
00:30:13.380 --> 00:30:17.750
Krzysztof Paruch: Cool. Okay, so, before we…

179
00:30:18.240 --> 00:30:23.899
Krzysztof Paruch: look into the framework and the standard. We actually want to

180
00:30:24.110 --> 00:30:31.069
Krzysztof Paruch: double-click on the distinction, what Henson was proposing, and, like, how he was looking into this 2003, 2007.

181
00:30:31.490 --> 00:30:49.800
Krzysztof Paruch: And how he was solving it economically. So, he addresses the same combinatorial problem on a different layer. He does not begin with the technical question of how claims should be tokenized. Back then, like, we didn't even have blockchains, like, we didn't have tokens, like, it was not feasible to talk about those things this way.

182
00:30:49.800 --> 00:31:05.849
Krzysztof Paruch: he was talking about centralized implementation of all of this, or kind of like some database solutions. But, he… he basically asked a simpler question, a more straightforward question, which is, how can a very large family of contingent claims be priced, updated, and kept internally consistent?

183
00:31:06.920 --> 00:31:08.920
Krzysztof Paruch: So,

184
00:31:09.230 --> 00:31:19.059
Krzysztof Paruch: So he… he proposed a thing that we've discussed, in multiple previous conversations already, which is the market scoring rule.

185
00:31:19.200 --> 00:31:28.509
Krzysztof Paruch: Which is, basically an aggregate scoring rule for the full market, a shared scoring rule with, for all participants, where

186
00:31:28.960 --> 00:31:35.720
Krzysztof Paruch: A change in the state is an update of beliefs, and the price you pay is the price for the change of inventory.

187
00:31:36.490 --> 00:31:45.900
Krzysztof Paruch: And then this was, this was his, his approach to this problem, and he then showed that, like, by choosing, good,

188
00:31:46.080 --> 00:31:53.220
Krzysztof Paruch: functions to derive the cost function, from, the market scoring rule from the cost function.

189
00:31:54.030 --> 00:32:04.480
Krzysztof Paruch: you can get the properties that you're looking for, and he found that logarithmic market scoring rules have all of those modularity properties that he was actually super interested in.

190
00:32:04.480 --> 00:32:21.980
Krzysztof Paruch: The ones that allow you to make those partitions of claims and make sure that the update of the leaves is only affecting the events that are actually affected by the contingency that is

stated in the particular statement.

191
00:32:21.980 --> 00:32:26.689
Krzysztof Paruch: And it's leaving other, probabilities unaffected.

192
00:32:26.690 --> 00:32:38.739
Krzysztof Paruch: And we want to talk this through briefly now. So, what we… how does this work? Yeah, we talked about this, but we have basically a belief state of P, a probability distribution.

193
00:32:38.740 --> 00:32:48.629
Krzysztof Paruch: of all outcomes, and we are moving it to P prime, and the system charges the implied cost of this update through a cost function, which is derived from market growing rules.

194
00:32:48.660 --> 00:33:02.389
Krzysztof Paruch: The market maker is maintaining this convex function C, that is mapping the outstanding vector of shares to the cumulative collateral required to support those positions, and what you pay is the difference,

195
00:33:02.390 --> 00:33:21.410
Krzysztof Paruch: in kind of, like, the position that you're moving the inventory to versus the inventory at which it was before you moved, so you're basically paying delta L, which is the cost function evaluated at S plus the delta S that you choose to move it to, minus the original state that it was in.

196
00:33:22.640 --> 00:33:33.140
Krzysztof Paruch: And, the prices, the marginal prices, the instantaneous prices, are derived from partial derivatives of this cost function in all of the shared directions.

197
00:33:33.650 --> 00:33:45.660
Krzysztof Paruch: all of the claim directions, and then you get, basically, to the price vector, which is Nabler of the cost function, and you get all of those P's, and those P's are normalized and represent probabilities.

198
00:33:47.330 --> 00:33:52.099
Krzysztof Paruch: And they tell you how much it costs you to purchase one additional unit of share, given the current state.

199
00:33:52.170 --> 00:33:53.600
Krzysztof Paruch: So, so this is…

200
00:33:53.600 --> 00:34:15.079
Krzysztof Paruch: this is what's happening here, and his logic is to trade beliefs rather than tokens, like, he basically asks you, okay, this is the inventory now, you want to move it to this inventory, let's evaluate the cost function here, let's evaluate it here, let's subtract it, and then I will tell you how much you have to pay for this move, and then you will be able to hold those claims, and you will be able to pay out.

201
00:34:15.080 --> 00:34:29.320
Krzysztof Paruch: And there are some properties of this. We mentioned that there is the bounded loss property of market scoring rules, where there is something that you need to subsidize it with, but there is kind of, like, a limit that is the maximum loss.

202
00:34:29.320 --> 00:34:40.629
Krzysztof Paruch: And so the market operator or the market initiator has to endow the market scoring rule with some capital, just to make sure that all claims can be actually redeemed.

203
00:34:40.630 --> 00:34:51.030
Krzysztof Paruch: But this is just in the worst-case scenario, this is not necessarily the amount that has to be paid out. But all the other collateral that's coming in is coming in from exactly those data updates from trading.

204
00:34:51.440 --> 00:34:56.270
Krzysztof Paruch: And then, at the end, there is sufficient capital, and the market maker is solvent.

205
00:34:56.920 --> 00:35:21.870
Krzysztof Paruch: Okay, so this is what's happening here, and we are not creating separate markets for every claim, but we are creating coherent market prices that… claims that derive from the joint distribution. And this is the thing. So, like, Hansen describes it economically. He says, like, you don't actually need to set up all of those distinct events, outcome tokens, and represent all of the claims.

206
00:35:21.870 --> 00:35:42.940
Krzysztof Paruch: Because the only thing I will be asking you about is kind of, like, the current belief across the full market state, and you will only need to pay the difference between the state that it was in and the state that you want to change it to. So, the cost function will tell you how much you have to pay. We don't really need to kind of, like, have all of those representations.

207
00:35:45.160 --> 00:35:54.580
Krzysztof Paruch: But in this paper, and you can look into this, again, I mentioned in the notes leading towards the meeting that we will not really look into the paper, but you can obviously look into it.

208
00:35:54.580 --> 00:36:06.120
Krzysztof Paruch: You will see that, there are things, described here in very much detail, and that you can, also, this,

209
00:36:06.190 --> 00:36:09.220
Krzysztof Paruch: See the limitations of this approach.

210
00:36:09.300 --> 00:36:16.900
Krzysztof Paruch: So there are, one of the limitations that he's mentioning, is,

211
00:36:17.180 --> 00:36:32.019
Krzysztof Paruch: that it's still, the computational complexity, and it's actually not really solving it, it fully. So if you have two large state spaces, there is still kind of, like, not, not, not full, redemption.

212
00:36:32.190 --> 00:36:43.819
Krzysztof Paruch: And, he discusses a few workarounds to solve, kind of, like, this complexity in a different fashion, which is base net representations, sparse dependency structures, overlapping local market makers.

213
00:36:43.820 --> 00:37:06.649
Krzysztof Paruch: and approximate inference methods. I have not looked into this at all, so I cannot tell you much about this. But, kind of, like, this is the storyline here. So, like, Hansen says, okay, we don't want to get into the details of representing each claim

individually, we want to talk about it on a different level, just a joint distribution, tell me the state updates, I will tell you the price, that's it.

214
00:37:06.650 --> 00:37:20.279
Krzysztof Paruch: Right? Okay, so what we see now is that the conditional token framework solves a different problem, right? So, like, we're not talking about the economic pricing architecture, we're talking about the underlying blockchain asset structure and the representation of the claims themselves.

215
00:37:20.280 --> 00:37:38.919
Krzysztof Paruch: So, we are now asking how claims can be represented, issued, transferred, recombined, and settled efficiently on-chain. So now we got there where we wanted to be, right? We've got all of the motivation out of the way, and now we can actually talk about what an ESC 1155 is, because this is the standard that is used.

216
00:37:39.120 --> 00:37:51.489
Krzysztof Paruch: To represent those claims. So, what is it? We've started here with an ERC20. We have one smart contract that represents fungible tokens. We have, one contract, one asset.

217
00:37:51.680 --> 00:38:09.699
Krzysztof Paruch: There is something in between, between ERC20 and ERC 1155, which is the ERC 721, where we have one contract that manages non-fungible tokens, so we have many unique assets. This works well for, for example, collections of different…

218
00:38:09.940 --> 00:38:26.940
Krzysztof Paruch: of art, right? Like, of NFT collections. Like, this has been very popular in this setup, where there was, one… one ERC721, for example, apes, and then you had, like, different, unique assets, like, different types of apes in…

219
00:38:26.940 --> 00:38:31.629
Krzysztof Paruch: In this contract, but, each of them was,

220
00:38:32.540 --> 00:38:49.379
Krzysztof Paruch: was set up individually. And now with the ESC 1155, we have an even more sophisticated or complicated structure, where we have one single contract that is able to manage simultaneously.

221
00:38:49.380 --> 00:38:56.880
Krzysztof Paruch: all the different kind of things. So, we have fungible tokens, semi-fungible tokens, unique items, arbitrary token classes, and

222
00:38:56.970 --> 00:39:12.490
Krzysztof Paruch: the whole idea behind it is that we have IDs within this ERC 1155, and all of those IDs are then treated separately, and you can talk about balances within this ID, so you can think of it

223
00:39:12.490 --> 00:39:29.040
Krzysztof Paruch: the ERC1155 to be something similar to a wrapper of multiple ERC20 tokens, for example. You can think of it this way. But it's even more… it's even richer than that, but this is a good mental model for what we need, at least, yeah.

224
00:39:29.920 --> 00:39:49.029
Krzysztof Paruch: This is mostly used in gaming and NFT ecosystems, because one contract has to contain currencies, weapons, tickets, skins, whatever is represented with this character, or with this level, or with this game at this moment, and those tokens have been carrying all of this information in its state, but also in its metadata.

225
00:39:49.660 --> 00:40:00.589
Krzysztof Paruch: So, this is exactly what we use, or what we need as a building block for our production market, because we have many related asset types, and we have those different claims.

226
00:40:00.590 --> 00:40:19.989
Krzysztof Paruch: And different combinations of them. So we want to have something that represents a yes claim on event A, which would be one of the IDs, then a no claim of event A. We have something, like, that represents candidate B wins, or maybe that is candidate A or B wins. So all of those things can be represented with different IDs, and then we can look at those balances of those IDs there.

227
00:40:20.550 --> 00:40:39.609
Krzysztof Paruch: So, if we had an ESC20, we would need a separate token contract, but we want to avoid this, so we are using an ESC 1155, and this is sold by one contract that hosts all of the claims. Each claim becomes a token ID. The balances within those claims remain fungible.

228
00:40:40.010 --> 00:40:56.430
Krzysztof Paruch: We can, batch, transfers, so we don't have always to go back and kind of, like, trade claims, but we can, within the token ID, also make some, some matching and some transferring and some, some, mutations.

229
00:40:56.880 --> 00:41:03.309
Krzysztof Paruch: And, we don't… we only need one contract, so we don't need multiple contracts per market.

230
00:41:04.730 --> 00:41:05.620
Krzysztof Paruch: Okay.

231
00:41:06.220 --> 00:41:10.129
Krzysztof Paruch: So, now… The final step would be…

232
00:41:10.380 --> 00:41:21.749
Krzysztof Paruch: the conceptual architecture. So let's look into the code of this CFT token, and what we can do here is we can look actually into diagnosis

233
00:41:21.860 --> 00:41:28.229
Krzysztof Paruch: GitHub, And we have the market makers, but we also have the token contracts.

234
00:41:28.350 --> 00:41:42.179
Krzysztof Paruch: So we have the condition token contract, and we can look at the conditional token soul contract. That is exactly the one that we want to discuss, and we want to see how… what is in there, which functions we have, and how they are used.

235
00:41:43.920 --> 00:41:53.880
Krzysztof Paruch: First, we talked briefly about the primitives. So we have conditions, which is basically the E that we had initially in the state space definition, the event.

236
00:41:54.040 --> 00:41:59.220
Krzysztof Paruch: We have the outcome slots, which were the A's, the mutually exclusive outcomes of the question.

237
00:41:59.330 --> 00:42:17.700
Krzysztof Paruch: We have collections, which are subset of those outcomes, so, like, the partitions of the event space, so to say. We have positions, which is if we mint claims, actually, by injecting capital or collateral into the contract, and then creating a position.

238
00:42:18.040 --> 00:42:35.189
Krzysztof Paruch: We have the token ID, which is the internal representation of the position within the contract, and we have a payout vector, which is determined by the oracle at the end of the lifetime of the contract that is telling us basically how much we have to pay out.

239
00:42:35.450 --> 00:42:39.320
Krzysztof Paruch: To each of the outcome shares.

240
00:42:40.020 --> 00:42:47.350
Krzysztof Paruch: And to, getting a little bit back to your question, Jamshit, I mean, when I…

241
00:42:47.460 --> 00:42:54.319
Krzysztof Paruch: I stumbled here across the definition, or the discussion, of whether

242
00:42:54.660 --> 00:43:13.489
Krzysztof Paruch: we would have a scenario where fractional payouts could also be possible, and the answer is that yes, in this setup, we can do this. Like, we don't actually have to have, exclusive events, to some extent. So, like, we can build more richer use cases than either A or B happens.

243
00:43:13.490 --> 00:43:22.819
Krzysztof Paruch: Because we can set up this in a way where we would have a payout vector that is, maybe 50-50, or 70-30, or,

244
00:43:23.090 --> 00:43:33.650
Krzysztof Paruch: even in, kind of, we don't need to create a particular claim for this event, we can even have a payout vector that allows us to pay out distinctively between multiple claims, so…

245
00:43:34.040 --> 00:43:43.370

Krzysztof Paruch: Loosen… loosening up a little bit the aerod de Bro property, where you have only one asset being paid out, one and the others being zero.

246
00:43:45.820 --> 00:44:00.130
Krzysztof Paruch: Good. So let's look at the lifecycle of the position. How does it look technically? We are looking both at the settlement layer, which is on the CTF level, and we're at the market trade layer.

247
00:44:00.130 --> 00:44:07.890
Krzysztof Paruch: So, we create the position, so we define the question, so to say, we define the outcomes.

248
00:44:08.020 --> 00:44:19.880
Krzysztof Paruch: This is happening through a function in the contract, which is called prepareCondition, and we define the oracle, the question identifier, and the number of outcome slots. So, basically, like, what is it? What is the object that we're dealing with?

249
00:44:20.020 --> 00:44:27.969
Krzysztof Paruch: Then, the next step is the collateral that is associated with the position, where we can split it into claims, where we have the function split position.

250
00:44:28.400 --> 00:44:38.109
Krzysztof Paruch: And this basically means, give me one unit of collateral, and I will give you, N tokens, like, one… a full vector of outcome shares.

251
00:44:39.160 --> 00:44:55.469
Krzysztof Paruch: Those claims, then, are represented as those token IDs within the year C1155, and they are credited to the user. So this is what happened, like, okay, if you give me one collateral, I give you a credit of each of the outcome shares, and I'm storing this for you.

252
00:44:55.650 --> 00:44:57.500
Krzysztof Paruch: Once you have this, you…

253
00:44:57.830 --> 00:45:12.299

Krzysztof Paruch: the next step is basically, okay, you have now those shares, like, now let's bring them to market, and let's do all of the price discovery and exchange. This is obviously not happening on the CTF level, but one level above.

254
00:45:12.310 --> 00:45:19.830
Krzysztof Paruch: Because the CTF does not determine prices, it does not trade, it only tracks balances and ownership. This is all what it does.

255
00:45:19.850 --> 00:45:38.270
Krzysztof Paruch: But if you want to look at, kind of, like, the market, you have to look into the markets that are distinct to contracts, also in the GitHub repo, that give you both options, implementations for a fixed product market maker, which is an AMM-style implementation, based on the conditional token framework.

256
00:45:38.670 --> 00:45:40.920
Krzysztof Paruch: And, and,

257
00:45:41.830 --> 00:45:50.689
Krzysztof Paruch: logarithmic market scoring over MarketMaker. So, Knosis provides both options in the repo, where you can say, now that you have those tokens.

258
00:45:50.900 --> 00:45:53.969
Krzysztof Paruch: Let me show you how you could find prices.

259
00:45:54.170 --> 00:45:58.450
Krzysztof Paruch: If you want to do it this way, or if you want to do it the other way.

260
00:46:00.040 --> 00:46:14.729
Krzysztof Paruch: What you always can do through the lifecycle of a token is the merge of claims. So you have this merge position thing, where whenever you have a full set of shares, you can merge it back to collateral, so, like, you don't have to wait until the resolution.

261
00:46:14.730 --> 00:46:22.200
Krzysztof Paruch: You can just say, okay, let me just redeem the full set immediately for one unit of collateral. You can exit the position.

262
00:46:22.440 --> 00:46:24.130
Krzysztof Paruch: And if you don't do it.

263
00:46:24.180 --> 00:46:39.760
Krzysztof Paruch: kind of like all the positions that you have are active until the end, and then the resolution comes where you would have the report payouts that is, called by the oracle. It reports the payout vector that determines how much, how much you want to

264
00:46:39.760 --> 00:46:56.589
Krzysztof Paruch: it has to pay to each of the respective claims. Fractional payouts are possible, and then what happens is that the balances still exist, so they are kind of, like, still on the representation layer, they are there. There is also this vector with payouts.

265
00:46:56.590 --> 00:47:08.370
Krzysztof Paruch: But not… and hence, we now know the redemption values of those shares, right? And only when, the user decides to redeem positions, so someone is calling him or

266
00:47:08.370 --> 00:47:16.920
Krzysztof Paruch: Like, anyone is calling the redeem position function function, then the claims are burned, liquidated, and the collateral is released to the owner.

267
00:47:17.120 --> 00:47:18.400
Krzysztof Paruch: of 2000.

268
00:47:19.640 --> 00:47:33.059
Krzysztof Paruch: Okay, so, when you look at it from… through the economic lens, what you see now is that initially you have collateral, which is the unconditional value, so, like, you have some money, like, you just have it, it's worth 100.

269
00:47:33.230 --> 00:47:48.179
Krzysztof Paruch: And then you use this money, or this asset, to fund claims, to create positions that are conditional, because you mint a full set of shares, like, for example, out of this 100 money, you mint 100 yes and 100 no shares.

270

00:47:48.180 --> 00:48:00.680
Krzysztof Paruch: And now, it's worth the same, but, not really, right? Because, like, they are conditioned claims. So only in aggregate, this is worth 100, because now if you start trading, and if you don't own

271
00:48:00.680 --> 00:48:12.619
Krzysztof Paruch: this full set at the end, it might, be worth zero as well, right? Like, if you sell all yes tokens, and yes is the outcome, then, like, your no shares are worthless.

272
00:48:12.920 --> 00:48:14.139
Krzysztof Paruch: And then.

273
00:48:14.490 --> 00:48:32.950
Krzysztof Paruch: at the end, you have the redeem payout, which is the realized value, so only if the outcome resolution is known, then those shares are actually worth something, you know exactly how much, and then you can do something with it. So this is what happens economically through the lifecycle of it.

274
00:48:33.980 --> 00:48:48.919
Krzysztof Paruch: Perfect! So this is the conditional token framework. This is it. So, like, this is how, using an ERC1155, you can now represent all of those, claims, those claim combinations, and you can do it more elegantly than if you were doing it

275
00:48:48.920 --> 00:48:59.289
Krzysztof Paruch: through an ERC20 token. So now the final question that remains, and obviously we can stop now and have a brief discussion, but the final question that remains at this stage is.

276
00:48:59.290 --> 00:49:10.820
Krzysztof Paruch: Now, how does it work on… on the market layer? So, how do you actually find prices for those claims, and how do you… how do you facilitate exchange, and… and what's happening here? How do you…

277
00:49:11.000 --> 00:49:13.480
Krzysztof Paruch: Yeah, act with those claims.

278
00:49:23.930 --> 00:49:25.200

Krzysztof Paruch: Comments, questions?

279
00:49:32.710 --> 00:49:45.519
Jamsheed Shorish: I'll open, of course, the floor up to other people who have more experience with either AMM design or even AMM research than I do, but I'm wondering, just from the perspective of how Gnosis was able to put all of this together, is this something that is

280
00:49:45.730 --> 00:50:03.200
Jamsheed Shorish: more or less replicated in other solutions as well? Is it, you know, publicly enough that basically anyone can spin off their own version of this? Or is there anything specific about it that affords them an advantage, let's say, in the marketplace for actually building prediction markets on top of their token architecture?

281
00:50:04.660 --> 00:50:23.020
Krzysztof Paruch: I don't know, like, everything about the history of how this came about, but what I can tell you is that PolyMarket is built on top of the conditional token framework, and basically everyone that you talk to right now is basically using it as the standard to develop their prediction markets. So, the answer is.

282
00:50:23.020 --> 00:50:29.039
Krzysztof Paruch: It appears, like, there appear to be consensus about this to be a proper representation

283
00:50:29.090 --> 00:50:40.500
Krzysztof Paruch: of the securities that we wanted to talk about. So it, and there are two options here, from my view. Like, option number one is actually the best representation, and it's

284
00:50:40.570 --> 00:50:48.279
Krzysztof Paruch: The thing to, like, do to, like, the proper, representation of those objects that we want to trade with.

285
00:50:48.510 --> 00:50:55.450
Krzysztof Paruch: or B, it's good enough, and people just started building on top of it, rather than…

286
00:50:55.650 --> 00:51:13.740
Krzysztof Paruch: trying to reinvent the wheel, and maybe improving it. So, I had a discussion with someone who says, okay, I'm

challenging this representation, there might be something that is better, and they are trying to develop something new, but the majority of current projects are taking this as given.

287
00:51:16.960 --> 00:51:23.220
Jamsheed Shorish: Gotcha. So it basically acts sort of as the underlying technology substrate for building these markets on top of. Okay.

288
00:51:23.220 --> 00:51:24.550
Krzysztof Paruch: Right. Yeah.

289
00:51:25.530 --> 00:51:26.210
Krzysztof Paruch: Right.

290
00:51:26.210 --> 00:51:26.630
Jamsheed Shorish: Thanks.

291
00:51:28.420 --> 00:51:30.679
Krzysztof Paruch: Cool. Any other questions, comments?

292
00:51:30.900 --> 00:51:39.650
David Sisson: Yeah, is there anything in here that… like, in the properties of AMMs and… Frankly, I…

293
00:51:39.850 --> 00:51:46.530
David Sisson: Particularly around where, sort of the effective liquidity on, on, on, on,

294
00:51:46.690 --> 00:51:53.049
David Sisson: Sort of the effect of withdrawing and inserting or adding coins.

295
00:51:53.260 --> 00:51:59.989
David Sisson: That addresses… Some of the risk, you know, around…

296
00:52:00.220 --> 00:52:10.970
David Sisson: gaming the markets. That, you know, you have a… you're trying to expose information, but you're doing so in a way that

297
00:52:11.150 --> 00:52:17.729
David Sisson: In the right conditions, somebody with insider information can make out like a bandit.

298
00:52:18.220 --> 00:52:21.929
David Sisson: Is there… Is there anything…

299
00:52:22.590 --> 00:52:24.759
David Sisson: On that one in particular, I'm thinking…

300
00:52:24.930 --> 00:52:32.249
David Sisson: you know, if you have a low liquidity AMM, and you put a whole bunch in, and you take a whole bunch out, you don't necessarily…

301
00:52:33.310 --> 00:52:35.250
David Sisson: get anywhere.

302
00:52:36.100 --> 00:52:54.539
Krzysztof Paruch: Right. Are you talking about, kind of, like, the particular limitations or the failure modes of one particular representation of such a market or such a claim? Or are you talking more about the performance across different venues? If you have one, kind of, like, designed like this, and the other like this, and you have both,

303
00:52:54.540 --> 00:52:56.449
Krzysztof Paruch: Tradable, actually, at the same time.

304
00:52:56.750 --> 00:53:08.190
David Sisson: I'm thinking both cases, but the… in my… in my, you know, the only, the only experience I really have is in the… in the individual case, you know, if,

305
00:53:08.800 --> 00:53:16.080
David Sisson: That, the fact that if you have a low liquidity market,

306
00:53:16.370 --> 00:53:23.570
David Sisson: A particular large move will… will have significant effect on the… on the… on the…

307
00:53:24.070 --> 00:53:28.029
David Sisson: On the value, not necessarily in a way that's beneficial to you.

308
00:53:29.700 --> 00:53:46.230
Krzysztof Paruch: Right. So there is nothing that I have found so far, and kind of, like, the representation of those claims that would have any impact on things that you mentioned here, except for the things that we've discussed in previous conversations, where we said.

309
00:53:46.270 --> 00:53:53.559
Krzysztof Paruch: Depending on the endowment with liquidity, and the parametrization, and the choice of the market scoring rule.

310
00:53:53.630 --> 00:54:02.990
Krzysztof Paruch: Also, more generally, the choice of the market microstructure, we will have different performance of, the prices.

311
00:54:03.380 --> 00:54:09.610
Krzysztof Paruch: As they are derived by those markets,

312
00:54:09.790 --> 00:54:21.280
Krzysztof Paruch: But not so much on, kind of, like, the asset definition itself, as we are discussing today. So, the answer is yes, obviously, there are, different,

313
00:54:21.960 --> 00:54:25.160
Krzysztof Paruch: Amplitudes, for example, of your effects.

314
00:54:25.390 --> 00:54:32.849
Krzysztof Paruch: If you, if you make a trade on one particular venue, if it's parameterized In some fashion.

315
00:54:33.750 --> 00:54:35.700
Krzysztof Paruch: But,

316
00:54:36.230 --> 00:54:43.420
Krzysztof Paruch: there is nothing that discusses, really, kind of, like, failure modes or economic attack. So, like, those properties are still…

317
00:54:44.100 --> 00:54:47.669
Krzysztof Paruch: Good enough, so to say, and that kind of, like, the…

318
00:54:48.680 --> 00:54:51.439
Krzysztof Paruch: The… the properties that you want to…

319
00:54:51.700 --> 00:54:58.710
Krzysztof Paruch: to have, they hold, so you don't have any economic exploits feasible with capital injections, for example, so, like, this is not possible.

320
00:54:59.420 --> 00:55:09.600
Krzysztof Paruch: But one thing that shows up here quite a bit is, the bounded loss property in the market scoring rule case. So, in…

321
00:55:09.730 --> 00:55:11.599
Krzysztof Paruch: In the AMM scenario.

322
00:55:12.080 --> 00:55:18.830
Krzysztof Paruch: There is, like, solvency per definition, but in the market scoring rule case, there is some…

323
00:55:19.300 --> 00:55:27.560
Krzysztof Paruch: endowment necessary. So the only way how I could see someone being in danger is the scenario where the implementation is

324
00:55:27.690 --> 00:55:30.180
Krzysztof Paruch: A market scoring rule, and…

325
00:55:30.390 --> 00:55:48.900
Krzysztof Paruch: the economic attack would be that you know much, much more than the person who is initializing it does, and therefore you can game the system by revealing information that is very beneficial to you and just collecting money, causing the other person to lose money. But this is part of the rules of the game in this case, so, yeah, I don't know, it's…

326
00:55:49.650 --> 00:55:54.599
Krzysztof Paruch: maybe, Jamsit, you know something that I'm missing

here, but I think that's… that's my point here.

327
00:55:55.700 --> 00:56:03.600
Jamsheed Shorish: I mean, it's evocative of insider trading, right? Essentially, if you're trying to open a new market for a new product, or even

328
00:56:03.690 --> 00:56:18.269
Jamsheed Shorish: the way in which we hope to have protections on IPOs is a way in which that regulation is set up to ensure that people aren't simply being fleeced because of a massive information asymmetry at the beginning.

329
00:56:18.340 --> 00:56:32.310
Jamsheed Shorish: So maybe there would have to be developed kind of a framework in which there would be some give around what the pricing would be, or a vesting period to make it detract from individuals sort of exploiting that at the beginning, until the…

330
00:56:32.340 --> 00:56:38.460
Jamsheed Shorish: Yeah, I don't… what you might call it, the variance sort of settles down after initialization has completed.

331
00:56:38.670 --> 00:56:44.829
Jamsheed Shorish: But, but yeah, without that, without those guards, you might find, you know, mispricing could be done.

332
00:56:44.980 --> 00:56:54.690
Jamsheed Shorish: Because you are actually working on things that maybe everyone collectively thinks are unknown, but maybe somebody has better information and can exploit.

333
00:56:56.110 --> 00:56:56.790
Krzysztof Paruch: Right.

334
00:56:57.190 --> 00:57:00.980
Krzysztof Paruch: Yeah, but this is the old problem that we have with all of those.

335
00:57:01.160 --> 00:57:20.760
Krzysztof Paruch: system security and market venues that, yeah, if

there is information asymmetry outside of the market, like, or at initialization, there's not much that you can do outside of regulation or governance or proper control of, like, why does this market exist in the first place, rather than how do you design it and how do you represent claims.

336
00:57:21.550 --> 00:57:27.379
Jamsheed Shorish: Well, you can… I mean, I'm thinking along the lines of… Basically, mitigating the…

337
00:57:27.590 --> 00:57:29.800
Jamsheed Shorish: Not so much mitigating their risk as sort of

338
00:57:30.070 --> 00:57:46.200
Jamsheed Shorish: cutting off the upside. If you can do that to the person who can make money off of the asymmetry, then you're sort of reducing the risk associated with being part of the system, both as the implementer, setting up the initialization and getting it wrong, but also of individuals who buy in to use it at the beginning.

339
00:57:46.200 --> 00:57:53.370
Jamsheed Shorish: And that's, of course, always the cliche about IPOs, right? The IPOs have this massively inflated value as the hype cycle reaches its peak, everybody jumps in.

340
00:57:53.370 --> 00:58:06.310
Jamsheed Shorish: you know, we're going to see this with, like, SpaceX soon, I guess. And then, of course, the price crashes, because people want to make that short-term profit, then the price drops down to some sort of fundamental value. And it's really the fundamental value that you want to predict, not so much the question of whether or not there's, like.

341
00:58:06.310 --> 00:58:13.990
Jamsheed Shorish: a juicy bit of information that sort of, like, kicks in at that point. So, I think it's still part of the design.

342
00:58:14.040 --> 00:58:33.219
Jamsheed Shorish: process, but as you say, it is relegated more along the lines of, like, okay, let's, using the governance system, kind of design the protocol such that there are these guards that are in place at the beginning, because everybody knows at the beginning, the initial conditions aren't, you know, aren't a spike. They aren't,

like, clustered around one sure thing, but are still very distributed.

343
00:58:34.790 --> 00:58:35.730
Krzysztof Paruch: Right.

344
00:58:36.450 --> 00:58:43.760
Krzysztof Paruch: Cool, perfect. Let me… give me the last 5 minutes to, like, complete the picture on the market layer. I think this is a good time spent to do this.

345
00:58:43.760 --> 00:59:08.680
Krzysztof Paruch: Just so that we kind of, like, also see, like, what does it mean if we want to actually discover prices for those assets? Or, like, how does Knosis propose to use those tokens in a market setup, right? So, CTF does not answer what is the current fair price, who is ready to buy and sell, we would need, basically, order books to achieve that, and how can traders enter or exit, how is liquidity pooled, all of this is not represented, but they provide two kind of

346
00:59:08.680 --> 00:59:25.469
Krzysztof Paruch: nautical automated market examples that I showed you already, which is the fixed product market maker, which is very akin to an AMM or DEX implementation that we've seen, or discussed in scenarios where we discussed Uniswap, and we have the logic market scoring rule market maker implementation as well.

347
00:59:25.470 --> 00:59:34.840
Krzysztof Paruch: So, how does the first of them work? Similar as AMMs work. So, like, it prices claims using pool inventories, so it has to hold,

348
00:59:34.840 --> 00:59:47.440
Krzysztof Paruch: balances of both claims, and then prices emerge from reserve balances rather than from an explicit probabilistic model, right? So if one token is scarce, then it increases in price.

349
00:59:47.440 --> 00:59:58.900
Krzysztof Paruch: And this is handled through the ERC 1155 standard, kind of, like, through those claim IDs that I mentioned before. So, like, not the full claim is traded, but kind of, like, balances of those claims are being traded and then put against each other.

350

00:59:59.020 --> 01:00:15.369
Krzysztof Paruch: So, how does this work? Collateral enters the pool, collateral is split into the complete set of outcome claims, yes or no, in the binary case. Then, the trader receives the desired outcome. So, if they only wanted to buy yes, they still have to pay, they enter the collateral, and they own, like.

351
01:00:15.370 --> 01:00:33.970
Krzysztof Paruch: a full set is minted, but the null token remains in the pool, and the yes token is handed over to the trader. So the pool retains the complementary claims, and the opposite trades can later be merged back into collateral. So, like, if someone sells back, then, like, it can be again redeemed, and the collateral can be again released.

352
01:00:34.200 --> 01:00:54.010
Krzysztof Paruch: In the logarithmic market scoring rule case, we have this approach followed by hands-on, where it maintains this cost function and just derives the price of moving the inventory, which is, again, here. So, like, we have those shares representation, the cost function, the price to move, the prices instantaneous are derived as such.

353
01:00:54.220 --> 01:01:12.909
Krzysztof Paruch: How this is represented in the code? In the constant function market maker, we have those, states stored, so we have, we are referencing to the conditional token framework, we have this ERC20 holdings, and we have the position IDs, and then, we hold,

354
01:01:12.910 --> 01:01:17.459
Krzysztof Paruch: inventories of those claimed tokens that are representative IDs. This is what's happening.

355
01:01:17.670 --> 01:01:29.299
Krzysztof Paruch: And, when the trader buys, exactly the process that I showed before happens. So, obviously, okay, there are some fees, but other than that, it's exactly the…

356
01:01:29.630 --> 01:01:43.350
Krzysztof Paruch: the process. How is it solved? The other way around. So, so, token is redeemed, it's combined with the complementary pool, then the, as the collateral is released, and, minus the fees.

357

01:01:44.550 --> 01:01:53.689
Krzysztof Paruch: How does it work on the logarithmic market side? Edtrader specifies how many claims to buy or to sell, and then the contract computes the implied cost of moving the inventory.

358
01:01:54.150 --> 01:02:05.599
Krzysztof Paruch: So, if we look at the trade lifecycle, we have a common starting point. The position is always this year C1155, with those claim IDs.

359
01:02:06.070 --> 01:02:07.439
Krzysztof Paruch: In both cases

360
01:02:07.550 --> 01:02:28.470
Krzysztof Paruch: If, at the market initialization, the liquidity provider sends collateral into the market, and then the contract receives the collateral, calls the split position, creates all of the full vector of shares, keeps only the claims and inventory that are unwanted, and gives the ones that are wanted to be bought to the buyer.

361
01:02:28.470 --> 01:02:30.700
Krzysztof Paruch: And minst those LP shares.

362
01:02:30.700 --> 01:02:34.929
Krzysztof Paruch: The pool holds yes or no claims, allows to trade.

363
01:02:34.960 --> 01:02:39.769
Krzysztof Paruch: And, in the next step, and then adjust the prices.

364
01:02:40.310 --> 01:02:44.310
Krzysztof Paruch: Here, the sponsor provides the funding, so this is the big difference.

365
01:02:44.570 --> 01:03:04.090
Krzysztof Paruch: This funding determines the parameterization, which is the liquidity sensitivity. So, like, depending on the choice of the scoring rule and the shape and the parameterization and the liquidity, you see exactly where you start and how your cost function is initialized, and then you have this initial cost state function that can be traded upon.

366
01:03:04.090 --> 01:03:13.550
Krzysztof Paruch: So, so what we are not talking about are token reserves, but we are talking about the point in this state space, the inventory that we're looking at.

367
01:03:13.790 --> 01:03:25.450
Krzysztof Paruch: And now, when trading comes in, the trader buys this position, sends collateral to the AMM, the AMM takes the fee, splits remaining collateral into the yes or no shares.

368
01:03:25.450 --> 01:03:34.659
Krzysztof Paruch: Transfers yes to the buyer, retains no in the inventory, and now you have the trader holding this yes as an ESC 1155 claim token.

369
01:03:36.030 --> 01:03:39.600
Krzysztof Paruch: And the price has changed because the balance has changed in the pool.

370
01:03:39.660 --> 01:03:56.210
Krzysztof Paruch: Here, if a trader buys yes, he basically states a quantity of delta Q, pays the price, and moves the inventory to this cost. So there is no change in inventory, but just moving the belief state. So this is what's happening here.

371
01:03:57.230 --> 01:04:09.210
Krzysztof Paruch: Then positions are held. They can be basically just on the token level again, on the conditional token framework level, they are held, or allocated, or mapped to the respective owner.

372
01:04:09.360 --> 01:04:11.570
Krzysztof Paruch: So this is what's happening here.

373
01:04:11.860 --> 01:04:25.669
Krzysztof Paruch: And then, when they sell back the position, the above process is reversed, here with the additional step that the collateral can be again withdrawn, because positions are merged, and then, like, the collateral is released.

374
01:04:25.960 --> 01:04:32.149
Krzysztof Paruch: Here, you are basically just inverting the

operation, so, like, you're moving back the inventory to the previous state at the cost.

375
01:04:33.350 --> 01:04:39.640
Krzysztof Paruch: At the end, the… the tokens are, like, when the outcome is known.

376
01:04:40.210 --> 01:04:51.350
Krzysztof Paruch: then the oracle just informs and pays out. So settlement is, again, on the CFD level, nothing happens. So this is basically the implementation that is provided by Knosis, and what…

377
01:04:51.400 --> 01:05:03.260
Krzysztof Paruch: Kind of, like, the outcome of the session maybe is, okay, we have found a way how we can represent very rich claims, and we have two candidate implementations that allow us to, to find even prices for those claims.

378
01:05:04.920 --> 01:05:07.029
Krzysztof Paruch: And with that, we're at time.

379
01:05:13.140 --> 01:05:13.810
David Sisson: Good deal.

380
01:05:15.310 --> 01:05:17.460
Jamsheed Shorish: Yeah, any questions? Sorry, sorry, Debbie, go ahead.

381
01:05:20.200 --> 01:05:29.499
David Sisson: No, I just… I think at this point, it becomes then a question of It… is… is there…

382
01:05:30.520 --> 01:05:34.170
David Sisson: You know, within this sort of primary-secondary market system.

383
01:05:34.810 --> 01:05:41.330
David Sisson: If the whole idea is… is to… Create a predictive signal.

384
01:05:42.100 --> 01:05:47.019
David Sisson: The next question is, how is that signal exposed?

385
01:05:47.760 --> 01:05:53.420
David Sisson: You know, is there a fire hose, market fire hose, that somebody can subscribe to?

386
01:05:53.630 --> 01:05:57.960
David Sisson: How… how… how is… is that part of what you're working on?

387
01:05:58.720 --> 01:06:11.449
Krzysztof Paruch: I mean, you can pull this data from APIs, obviously, so, like, you can connect to the data feed of the market, and obviously you can also observe it through dashboards, so, like, those mark… those prices are public, right? So, like, you can… you can just see…

388
01:06:11.450 --> 01:06:27.039
Krzysztof Paruch: at which price you would be able to see… to trade particular assets or particular claims. And through normalization, you just derive that those are the probabilities of those events. So, like, you have two information. Immediately, you say, like, this is how much… this is the price I need to pay to participate in the marketplace.

389
01:06:27.240 --> 01:06:33.349
Krzysztof Paruch: And point B is this is the aggregate market belief of this event being realized in the future.

390
01:06:46.300 --> 01:06:59.150
Jamsheed Shorish: Nice. Well, it seems like it is… it is at least ensuring that you can figure out a way to… to get prices to reveal information insofar as it's actually changing the probabilities. Of course, we had that discussion in the past

391
01:06:59.150 --> 01:07:08.180
Jamsheed Shorish: I don't want to take too much time with it, but just with regard to the notion of subjective probabilities you carry and hold in your head versus the aggregate probabilities that are reflected in the empirical frequency.

392
01:07:08.180 --> 01:07:21.089
Jamsheed Shorish: I think it's partially mitigated here, because you

do have that ability to see the scoring rule, and you know, kind of, the cost function, because it's all publicly information… public information that people are conditioning on when they're deciding how much to pay.

393
01:07:21.090 --> 01:07:39.299
Jamsheed Shorish: But again, I'm not so sure that it gives you exactly what you're looking for in terms of what a prediction market is supposed to be able to generate with regard to subjective beliefs. So that particular question that we raised a couple weeks back may still persist, but I agree, I think it seems a lot more likely that it's being addressed in this framework.

394
01:07:42.780 --> 01:07:59.029
Krzysztof Paruch: Exactly. So, for me, this was very nice, kind of, like, discovering, okay, like, because I was kind of ignoring the implementation standard up until now, and I was naively assuming ERC20 is enough, but, it was very refreshing to discover that there is actually a very

395
01:07:59.550 --> 01:08:08.729
Krzysztof Paruch: sophisticated and well-developed standard already in the pipeline that can be leveraged to actually realize, Henson's proposition from 23 years ago.

396
01:08:11.590 --> 01:08:12.030
Jamsheed Shorish: Cool.

397
01:08:12.030 --> 01:08:12.570
Krzysztof Paruch: Perfect.

398
01:08:12.800 --> 01:08:21.290
Krzysztof Paruch: So let's see where this journey will spring next time. I think we have the next meeting in roughly 4 weeks. There are a few topics, again, that are already now candidates.

399
01:08:21.290 --> 01:08:39.989
Krzysztof Paruch: We will see what makes sense. I'm looking to explore the TraderViewer at some point, just to see, kind of, like, how this connects, not only the market designer's perspective, but we will get there, so I will keep you informed. I will share the notes in the Slack channel, as always, and again, thank you so much for the participation and the very constructive feedback and discussions

today.

400
01:08:40.410 --> 01:08:59.239
Jamsheed Shorish: Excellent. Well, thanks again, Chris, as always, and I'm kind of hoping that we get more, yeah, people actually kind of jump on board a little bit. I know there's some people who are kind of thinking about things that are similar. I think Hashir just dropped off, but I think he may be interested. So, yeah, let's… looking forward to see how the, how the follow-up continues. But, appreciate, as always, your, your time.

401
01:09:00.029 --> 01:09:00.590
David Sisson: Perfect.

402
01:09:00.930 --> 01:09:02.520
Krzysztof Paruch: See you soon. Bye-bye.

403
01:09:02.529 --> 01:09:03.989
Jamsheed Shorish: Good. Yep, see you.